vllm-sr/d3-nano

Like 26
Follow vLLM Semantic Router 410

Zero-Shot Classification · Transformers · Safetensors · qwen3_5 · feature-extraction · decision-model · classification · system-one · multimodal · vision · video · custom_code · License: apache-2.0

Deploy · Copy to bucket NEW · Use this model



d3-nano

d3-nano is the 2B multimodal foundation decision model of Decision 3.0, the decision models of vLLM Semantic Router. Give it an input (text or JSON, optionally with images and videos) and the questions you need answered: pick one of several options, say yes or no, or rate on a scale. It answers them all in one call and returns a probability for every answer, without generating text.

Parameters	2.21B, including the 0.33B vision encoder

Downloads last month
887

Safetensors

Model size | 2B params
Tensor type | BF16
Chat template
Files info

Inference Providers NEW

Zero-Shot Classification

This model isn't deployed by any Inference Provider.

Ask for provider support

Model tree for vllm-sr/d3-nano

Base model — Qwen/Qwen3.5-2B-Base
Finetuned — Qwen/Qwen3.5-2B
Finetuned (495) — this model
Finetunes — 2 models
Quantizations — 8 models

Inputs	Text or JSON, plus images and videos (several per request)
Decision types	Choice · Yes / No · Score
License	Apache-2.0

Highlights

- **Jev Decision Index 0.3.2: 38.9 on the official text board**, first among the entries of similar size and on the board's Pareto frontier; public suite 42.22, measured with the official 0.3 kit on the released weights: all 140,178 public requests answered, none unsupported.
- **+11.8 over Decision 2.0** (its 2B model: 27.1 on the official board): +12.2 on the public suite, ahead in all five of its areas, and +13.2 and +6.6 on the private same-skill tests and new-domain tasks.
- **Reads images:** multiple images per request (PNG, JPEG or WebP), given as paths, URLs, PIL images or base64 data URLs; every question of the request sees all of them.
- **Reads videos:** multiple videos per request (MP4, WebM, MOV or MKV), given as paths, URLs, base64 data URLs or frame arrays, read at 2 frames per second; **Perception Test (multiple-choice video QA): 63.5** (internal evaluation).
- **Speed:** a median of 13.4 ms for a text request, 90.2 ms for a request with an image and 301 ms for a request with a 10-second video, on one

Spaces using `vllm-sr/d3-nano` 3

- multimodalart/jev-decision-index
- tardellirs/model-pulse
- emanforhelp/model-pulse

Collection including `vllm-sr/d3-n…`

Decision 3.0 Collection
Towards O… • 7 items • Upd… • △ 97

AMD Instinct MI325X GPU, one request at a time with AutoModel.

- **Serving:** 318 requests/s with p99 under 1 s on one AMD Instinct MI325X with vLLM Semantic Router, 4.3× AutoModel (one-question text, `batching` profile); the default `exact` profile gives AutoModel's answers byte for byte when both use the same kernel settings (Serving performance).
- **Many questions, one call:** Choice, Yes / No and Score questions about the same input are answered together, each from its own forward pass over the input, with a probability for every option.

Quickstart

```
pip install "transformers==5.17.0" torch tor
pip install flash-linear-attention  # option
```

```
import json

from huggingface_hub import hf_hub_download
from transformers import AutoModel

model = AutoModel.from_pretrained("vllm-sr/d

# Text
result = model.system_one(
    state="The order arrived damaged yesterda
    questions={
        "route": {
            "type": "choice",
            "instructions": "Which team shou
            "criteria": {
                "returns": "Refunds, replace
```

```
                "billing": "Payments, invoic
                "technical": "Product setup a
            }
        },
        "receipt": {
            "type": "noul",
            "instructions": "Does the custom
        },
        "urgency": {
            "type": "score",
            "instructions": "How urgent is th
            "criteria": [
                "Routine",
                "Soon",
                "Today"
            ]
        }
    },
)
print(json.dumps(result["answers"], indent=2)

# Text and an image (or several)
receipt = hf_hub_download("vllm-sr/d3-nano",
result = model.system_one(
    state="The customer says the blender arri
    images=[receipt],  # local paths, http(s
    questions={
        "route": {
            "type": "choice",
            "instructions": "Which team shoul
            "criteria": {
                "returns": "Refunds, replacem
                "billing": "Payments, invoic
                "technical": "Product setup a
            }
        },
        "on_receipt": {
            "type": "noul",
            "instructions": "Does the receipt
        },
        "payment": {
```

```
            "type": "choice",
            "instructions": "How was the ord
            "criteria": {
                "card": None,
                "cash": None,
                "gift card": None
            }
        }
    },
)
print(json.dumps(result["answers"], indent=2

# Text and a video (or several)
clip = hf_hub_download("vllm-sr/d3-nano", "a
result = model.system_one(
    state="A short clip from a test camera."
    videos=[clip],  # local paths, http(s) U
    questions={
        "direction": {
            "type": "choice",
            "instructions": "Which way does
            "criteria": {
                "right": "From left to right
                "left": "From right to left"
                "still": "It does not move"
            }
        },
        "color_change": {
            "type": "noul",
            "instructions": "Does the square
        }
    },
)
print(json.dumps(result["answers"], indent=2

# Or as a pipeline:
# transformers.pipeline("decision", model="v
```

Images go before the text of the request, each read at up to 1.6 megapixels; videos follow the images,

read at 2 frames per second (at most 32 frames spread over the whole video, each at up to 0.2 megapixels). Every question of the request sees all of them.

Run with vLLM Semantic Router

vLLM Semantic Router serves Decision 3.0 natively in Engine mode, which runs one model behind the System One API on your own machine. Its answers match this package's own `d3_server.py` exactly, for text, images and videos (measured on AMD Instinct MI325X).

```
curl -fsSL https://vllm-sr.ai/install.sh | ba
export PATH="$HOME/.local/bin:$PATH"
vllm-sr serve -e vllm-sr/d3-nano
```

The platform is detected automatically; choose one with `--platform auto|cpu|cuda|rocm` (default `auto`).

Engine mode and Decision 3.0 are in the development channel (`--channel dev`) until a stable vLLM Semantic Router release includes them. The first start pulls the image for your machine and downloads the model. Then ask questions about a request:

```
curl -s localhost:8899/v1/systemone -H 'conte
  "model": "vllm-sr/d3-nano",
  "state": "Write a Python function that merg
  "questions": {
    "kind": {"type": "choice", "instructions
             "criteria": {"code": "Writing o
    "reasoning": {"type": "noul", "instructi
```

```
    }
  }'
```

Send images as base64 data URLs (PNG, JPEG or WebP) in `images` and videos (MP4, WebM, QuickTime or Matroska) in `videos`; every question sees them:

```
curl -s localhost:8899/v1/systemone -H 'conte
{
  "model": "vllm-sr/d3-nano",
  "state": "What does the attached image sho
  "images": ["$(base64
  "questions": {
    "kind": {"type": "choice", "instructions
             "criteria": {"chart": "A chart
  }
}
EOF
```

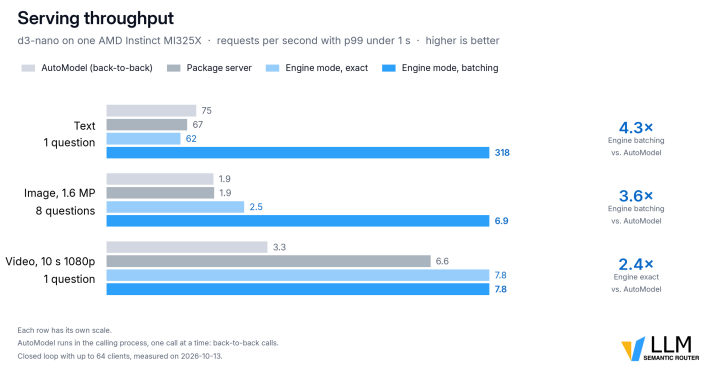
```
curl -s localhost:8899/v1/systemone -H 'conte
{
  "model": "vllm-sr/d3-nano",
  "state": "What happens in the clip?",
  "videos": ["data:video/mp4; base64,$(base64
  "questions": {
    "moving": {"type": "noul", "instructions
  }
}
EOF
```

`vllm-sr stop` stops it. See Router and Engine modes for limits and options.

Serving performance

On one AMD Instinct MI325X, vLLM Semantic Router serves d3-nano at 318 requests/s with p99 under 1

s, 4.3× AutoModel (one-question text, `batching` profile), and its default `exact` profile gives AutoModel's answers byte for byte when both use the same kernel settings.



AutoModel runs in your own process, one call at a time (the Quickstart code); the package server is its `python d3_server.py`; Engine mode is `vllm-sr serve -e` with its default `exact` profile, or with `--runtime-profile batching`.

Where Engine mode helps, and what it costs:

- **Throughput:** 4.3× AutoModel's requests per second on one-question text, 3.6× on 8-question images and 2.4× on 10-second videos, with the better Engine profile of each row.
- **Start-up:** the first answer comes 13.4 s after launch with empty kernel caches, against 25.9 s for AutoModel, whose first image then takes another 1.19 s, and 38.0 s for the package server.
- **Many questions and long inputs:** one request at a time, Engine mode (`exact`) answers 16 text questions in 54.2 ms (AutoModel: 57.7 ms), 8 questions about one image in 452 ms (519 ms) and a 32k-token text in 445 ms (477 ms); `batching` runs the 8 image questions together in 204 ms, 61 % less than AutoModel.

- **GPU memory:** 16.5 GiB per replica at peak with `exact`, about the same as the package server's 16.4 GiB, and 22.7 GiB with `batching`; AutoModel, serving one request at a time, peaked at 9.1 GiB. Engine mode keeps up to 2 GiB of image features cached.
- **Answers:** Engine mode (`exact`) returned answers byte-identical to AutoModel's on all 7 request types (text with 1, 4 or 16 questions, an image with 1 or 8, the video and a 32k-token text) when AutoModel used the same kernel settings: 140 Engine replies and 21 AutoModel calls compared. With the kernel autotuning AutoModel runs in each new process, 1 of the 7 was byte-identical and no decision changed (largest probability difference 0.011). `batching` changed 0.153 % of decisions on a 4,000-request sample of the public suite (45 of 29,339 questions, near-ties), with gold accuracy 87.707 → 87.744 %.

A single short request is fastest in your own process: AutoModel answers one text question 2.8 ms sooner than Engine mode and one new image 26 ms sooner, the cost of going through the router to the model worker and its media path. Engine mode's gains come from serving many requests at once.

▶ Single-request latency and the numbers behind the chart

Measured on 2026-10-13, one AMD Instinct MI325X per replica and each side alone on the GPU: ROCm 7.2 with PyTorch 2.12; AutoModel and the package server with Transformers 5.17.0; vLLM Semantic Router Engine mode built from source, with serving optimizations that are newer than its published images. Requests: a support ticket

of about 330 tokens with 1, 4 or 16 questions; a 1,456 × 1,092 PNG (1.6 MP) with 1 or 8 questions; a 10-second 1080p MP4, read at 2 frames per second; a 32k-token text; every image and video request carries new bytes. One request at a time: 3 warm-up calls, then up to 20 timed calls. Under load: 1, 2, 4, … 64 closed-loop clients, each level warmed up and then counted for 15 to 60 s; Engine mode admits 32 requests in flight per worker and answers the rest at once with 503. Start-up: model files on local disk; Engine mode from the start of its router process, without the CLI's image pull.

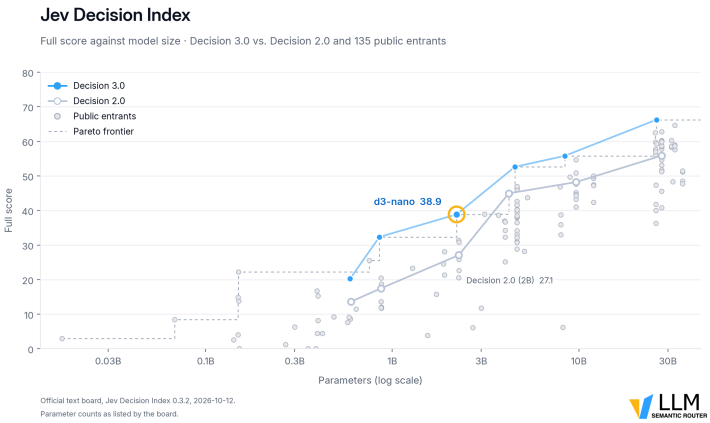
Evaluation

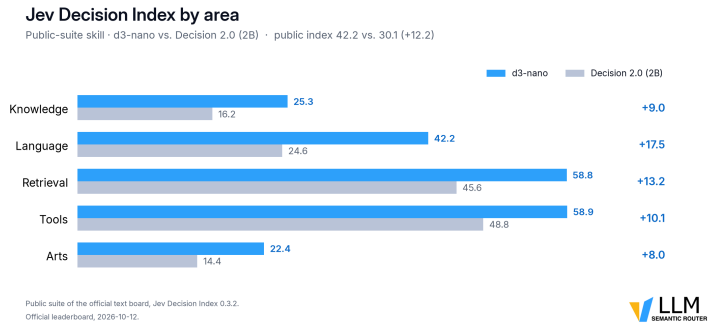
Text: Jev Decision Index 0.3.2

Model	Jev Decision Index ↑	Public ↑	Same-skill tests ↑	New-domain tasks ↑
d3-nano	**38.9**	**42.2**	**42.5**	**29.7**
Jev-Style 2B v3	31.4	34.1	34.6	25.2
Sieve 2B Plus	30.8	37.3	34.9	21.8
vera-core	28.1	30.1	29.1	25.7
Decision 2.0 (2B)	27.1	30.1	29.3	23.0
Decider 2B	25.8	29.8	28.5	21.0

On the official text board, d3-nano is on the Pareto frontier, where no entry with as many parameters or fewer scores higher, and ranks first among the 12 entries within 0.75 to 1.33 times its size. LiquidAI d1-3B, a larger model with 3.1 billion parameters, also

scores 38.9. Five of the six Decision 3.0 models are on that frontier; d3-edge, second among the entries of its size, is not.





Images: Jev Decision Index vision board 0.3.2

Model	Vision Index ↑	Public ↑	Private ↑
d3-nano	**54.7**	**59.2**	**50.2**
Intern-Decision-2B	47.8	52.9	42.7

On the official vision board, the six Decision 3.0 models form the whole Pareto frontier: no other entry is on it. d3-nano is ahead of Intern-Decision-2B, the only other entry within 0.75 to 1.33 times its size.

d3-nano on the public vision benchmarks († approximate rebuild):

Benchmark	d3-nano
CV-Bench	71.9
BLINK	49.9
RealWorldQA	53.7
CharXiv †	76.2
InfographicVQA †	86.9
Mind2Web †	66.2
Winoground	72.0
KIE (CORD+FUNSD) †	97.3
Moderation (Hateful Memes)	23.2
R-Bench-M	14.8
MMMU-Pro vision	18.6

Videos: Perception Test

Perception Test (validation)	**d3-nano**
All 19,140 questions	**63.5**
Memory	53.8
Abstraction	55.7
Physics	56.0
Semantics	81.0

Multiple-choice video question answering (three options per question, chance 33.3): top-1 accuracy (%), videos read with the defaults above.

Jev Decision Index: official leaderboard, text and vision boards 0.3.2, 2026-10-12; the boards evaluated v3.1.0, which has the same weights as this version. Size bands and the Pareto frontier use the parameter counts listed by the boards. d3-nano on the public vision benchmarks and the Perception Test: internal evaluation.

License

Apache-2.0 (LICENSE). Built on Qwen/Qwen3.5-2B (Apache-2.0).

Citation

```
@misc{d3_nano_2026,
  title        = {{d3-nano}: A Multimodal Fou
  author       = {{vLLM Semantic Router Team}
  year         = {2026},
  howpublished = {\url{https://huggingface.co
}
```

Trained on AMD Instinct MI325X GPUs.

System theme TOS Privacy About Careers Models Datasets Spaces Pricing Docs